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# Structurally Verifying Ecological Safety: An Empirically Grounded Framework for a Five-Layer Ecosafety Stack

## Establishing Empirically Grounded Kernels Across the Five Layers

The foundational objective of this research is to advance the `ecosafety-core-v2` specification by establishing a tightly coupled relationship between empirical calibration and formal verification . The initial phase of this work is dedicated to anchoring each of the five interdependent layers—hydraulics/FOG, biodegradable materials, biodiversity, carbon intensity, and data quality—with empirically grounded, physically plausible computational kernels. This step is critical; without it, any subsequent formal analysis would apply to an abstract model disconnected from reality, leading to fragile and potentially dangerous actuation . The goal is to move beyond theoretical constructs or "toy" models to produce calibrated parameters that reflect real-world conditions, thereby generating non-toy, physically plausible risk coordinates . Each layer requires a distinct but equally rigorous approach to calibration, leveraging the most appropriate scientific methodologies and data sources available.

For the hydraulics and FOG (Fats, Oils, and Grease) layer, the primary task is to establish kernels that accurately represent hydraulic loads, surcharge, and fouling potential. The calibration process begins with the application of the CEIM (Calibrated Environmental Impact Model) to historical Phoenix shards containing relevant metrics such as PFAS, E. coli, Total Dissolved Solids (TDS), and biodegradable node data . These systematic replays allow for the fitting of safe, gold, and hard corridor bands for key variables like energy use, hydraulic stress, biological load, and carbon impact . The resulting output is a fully populated `MetricField` catalog with defined `safemin/max`, `goldmin/max`, and `hardmin/max` values for each variable, which feeds directly into the `ecosafety.corridors.v2` schema . This process ensures that the hydraulic risk coordinates, such as rHLR (hydraulic loading rate) and rsurcharge, are not arbitrary but are derived from observed performance within a known environment. The calibration of the rFOG coordinate, in particular, hinges on matching predictions of surcharge and fouling from simulation models to actual field observations of FOG formation . This

empirical grounding is essential before proceeding to prove that any increase in these physical quantities monotonically increases their corresponding risk coordinate `r_j` and thus the global residual `V_t`.

The biodegradable materials layer requires a similar commitment to empirical validation, focusing on the kinetics of decomposition and its ecological consequences. The core of the calibration effort involves expanding the existing MaterialKinetics test bank with standardized laboratory runs that adhere to ISO 14851 and OECD 201/202/308 protocols . These tests must be conducted on recycled and biodegradable compounds under stressors representative of the target environment, such as elevated temperatures (40–55 °C), low relative humidity, dust exposure, and co-contamination with substances like PFAS . The output of these campaigns will be detailed time-series data on metrics like t90 (the time for 90% degradation), rtox (toxicity), rmicro (microbial residue), rCEC (cation exchange capacity), and PFAS residual content . This data serves to refine the `rmaterials` kernels and their associated corridor tables for specific substrate recipes, such as AntRecycling and FlowVac substrates . Furthermore, long-horizon decomposition simulations using tools like `decomposition.sim` will be run for these materials under realistic hydraulic and thermal conditions. These simulated results will then be validated against data collected from field trays and actual FlowVac deployments, creating a powerful feedback loop between laboratory science, simulation, and real-world performance . The ultimate goal is to confirm that for ker-deployable recipes, key decay metrics like rt90 and rtox never enter their "hard" corridor bands, providing proof data that can be used to certify the safety of these materials .

Calibration for the biodiversity layer is centered on quantifying landscape connectivity and ecological health. The methodology relies heavily on Geographic Information Systems (GIS) and advanced spatial modeling techniques. Connectivity indices, such as those derived from circuit theory and least-cost path (LCP) analysis, are used to quantify the degree to which a landscape facilitates or impedes movement among resource patches [6] [7]. Circuit theory, in particular, provides a robust framework for modeling organismal movement as a random walk across a resistance surface, offering a more realistic representation than single-optimal-path models [8] [9]. The calibration process involves using software like Linkage Mapper to simulate species migration and identify critical corridors and bottlenecks [7]. To ground these models in reality, they must be validated with empirical field data. This includes conducting field surveys around MAR (Marine Artificial Reef), reef, and wetland nodes to measure structural complexity, colonization rates, and patch importance . Key metrics like the Patch Importance Index (dPC) and the Delta Integral Index of Connectivity (dIIC) are calculated to identify ecologically critical patches [7]. The empirical ranges gathered from these field surveys

are then used to define the gold and hard corridor boundaries for the `rbiodiversity` metric, ensuring that the kernel reflects measurable ecological states rather than purely theoretical constructs . The validity of these models can be further tested by comparing predicted movement paths with observed trajectories from tracking studies, as demonstrated in research involving European hedgehogs [6].

For the carbon layer, calibration is based on life cycle assessment (LCA) principles and standards. The primary kernel parameter is the lifecycle COe (carbon dioxide equivalent) intensity, typically measured in kg COe per kWh or per operational cycle . Calibrating this layer involves drawing upon extensive, harmonized LCA datasets that provide global ranges of building material intensities, such as those found in the RASMI database [36]. These databases differentiate intensities by construction material, region, and building type, allowing for highly specific and context-aware corridor definitions [36][100]. The calibration process must adhere to internationally recognized standards like ISO 14040 and ISO 14044 to ensure consistency and comparability [118][127]. A comprehensive literature review and meta-analysis of existing LCA studies are necessary to clarify calculation methodologies and address disparities in inventory data, which can vary significantly even for engineered bio-based materials [34][39]. The resulting `rcarbon` corridors are tied to these lifecycle intensity thresholds, which may be benchmarked against regulatory requirements or best-in-class industry performance to define the safe, gold, and hard bands [33]. The use of established frameworks like the Digital Twin-Enabled Life Cycle Assessment Framework (DT-LCAF) can further enhance the accuracy of embodied carbon accounting during the construction phase [90].

Finally, the data quality plane, represented by the `rsigma` (uncertainty) and `rcalib` (calibration) coordinates, is calibrated based on metrological principles rather than environmental hazards. The goal is to create quantitative bounds on measurement error, sensor drift, and data ingest anomalies. The calibration of `rsigma` is informed by metrology and calibration evidence, specifically following the guidelines of international standards like ISO/IEC 17025 [1][2]. This standard provides requirements for the competence of testing and calibration laboratories and emphasizes the importance of confirming that a target measurement uncertainty can be met [50][60]. The process involves quantifying uncertainty using procedures consistent with the Guide to the Expression of Uncertainty in Measurement (GUM), often by linking sensor readings to reference instruments that have been calibrated by accredited bodies [1][4]. Studies comparing GUM methods with Monte Carlo simulations show that the choice of method can yield different uncertainty values, highlighting the need for a consistent and well-documented procedure [3][53]. The calibration of `rcalib` is achieved through a more direct experimental approach. By intentionally introducing controlled schema violations (e.g.,

missing columns, incorrect data types) and monitoring the impact on data ingestion, system residuals (`V_t`), and overall KER scores, a quantitative mapping can be established from ingest anomalies to increments in the `rcalib` coordinate . This allows the system to implement a strict "no schema, no write" policy, where deviations from the canonical `qpudatashard` schema trigger explicit penalties, mitigating the risk of bad data corrupting the entire ecosafety stack .

| Layer | Core Kernel Parameter(s) | Primary Calibration Methodology & Data Sources |
|---|---|---|
| **Hydraulics/ FOG** | Hydraulic Load (CEIM), Surcharge, Fouling Potential (rFOG) | Replay of historical Phoenix shards (PFAS, E. coli, TDS); Simulation with design codes; Field correlation of fouling predictions vs. observations [23] |
| **Materials** | Degradation Time (t90), Toxicity (rtox), Microbial Residue (rmicro), Leaching (rleachCEC) | Standardized lab tests (ISO 14851, OECD 201/202/308); Long-horizon simulations; Field validation of trays and deployed substrates [25] [128][129] |
| **Biodiversity** | Connectivity Index, Structural Complexity, Colonization Metrics (rbiodiversity) | GIS-based modeling (Circuit Theory, LCP); Field surveys of patch importance and colonization; Validation via animal tracking data [6] [7] [9] |
| **Carbon** | Lifecycle COe Intensity (kg COe/kWh or per cycle) | Harmonized LCA databases (RASMI); Adherence to ISO 14044 standards; Meta-analysis of published LCA studies [34] [36] [118] |
| **Data Quality** | Measurement Uncertainty (rsigma), Ingest Schema Violations (rcalib) | Metrological traceability (ISO/IEC 17025); GUM-compliant uncertainty propagation; Controlled injection of ingest errors to measure KER degradation [1] [2] [29] |

# Formal Verification of Shared System-Wide Invariants

Once the five-layer ecosafety stack has been anchored with empirically grounded kernels, the second and equally critical phase of the research framework is the rigorous formal verification of shared system-wide invariants . This phase moves beyond the question of whether the system *works* to the deeper question of why it *must* work safely. The objective is to construct watertight mathematical proofs that demonstrate the structural properties of the system, making harmful states unrepresentable by design . These proofs provide the ultimate assurance of safety and stability, transforming the ecosafety stack from a reactive monitor into a proactive, provably safe control system. The formal verification effort focuses on three interconnected pillars: proving the monotonicity of risk coordinates, establishing Lyapunov stability for the global residual, and guaranteeing compliance with the KER (Knowledge, Eco-impact, Risk-of-harm) governance framework.

The first pillar of formal verification is the proof of monotonicity for each risk coordinate, `r_j`. Monotonicity is the property that ensures an increase in the underlying physical or

ecological harm leads to a corresponding increase in the normalized risk value `r_j`. This is a fundamental requirement because it validates the very purpose of the corridor system. If tightening a corridor (i.e., shrinking the acceptable range for `r_j`) does not consistently lead to a reduction in the underlying hazard, the mechanism is meaningless. For example, if higher concentrations of a toxin were to map to a lower `rtox` value, the system would be dangerously inverted. Therefore, for every variable in the `RiskVector`—including `rFOG`, `rmaterials`, `rbiodiversity`, `rcarbon`, and `rcalib`—a formal proof must be constructed demonstrating that the normalization function is monotone increasing with respect to the harmful quantity it represents [1,12]. This proof typically involves analyzing the piecewise linear or other functional form used to map raw physical measurements (e.g., micrograms/liter of PFAS, days for 90% material degradation) to the unit interval `[0,1]`. The successful proof of monotonicity for all planes ensures that the entire system responds predictably and conservatively to increasing stress.

The second, and most crucial, pillar is the formal proof of Lyapunov stability for the global quadratic residual, defined as $V_t = \sum_j w_j r_j^2$ . A Lyapunov function is a scalar function used to prove the stability of an equilibrium point in a dynamical system. In this context, the goal is to prove that $V_t$ serves as a valid Lyapunov function for the ecosafety system. This requires demonstrating two main conditions: that $V_t$ is positive definite (which it is, by construction, as it is a weighted sum of squared terms with non-negative weights $w_j$), and that its change over time, $\Delta V_t = V_{t+1} - V_t$, is negative semi-definite under the system's operational rules [15,68]. The latter condition translates directly to the `safestep` invariant, which stipulates that for any valid action or transition allowed by the system, the new residual must satisfy $V_{t+1} \leq V_t$ . Proving this invariant is the core mathematical guarantee of system safety. It structurally prevents the accumulation of risk over time and ensures that the system is always moving towards a safer state or at least not a less safe one. The proof would assume the calibrated kernels and their monotonicity properties, then analyze the logic of the `safestep` function to show that it enforces this non-increase. Techniques from formal verification, such as those used to synthesize Lyapunov functions for nonlinear systems or to verify barrier functions, could be adapted for this purpose [62,64,69]. Successfully proving this invariant means that the entire multi-dimensional risk space is governed by a single, unifying stability principle.

The third pillar of formal verification is the proof of KER compliance. The KER framework—comprising Knowledge-factor (K), Eco-impact (E), and Risk-of-harm (R)—provides the practical, quantitative gates for deploying and promoting the system's capabilities . While the monotonicity and Lyapunov proofs establish the underlying physics of safety, the KER proofs connect these abstractions to concrete deployment criteria. The research plan specifies target thresholds for production lanes: K ⩾ 0.90, E ⩾ 0.90, and R ⩽ 0.13 . The

formal proof obligation here is to demonstrate that adherence to the corridor rules and the `kerdeployable` gate inherently leads to achieving these targets. This involves mathematically linking the structure of the corridor system (the existence of safe, gold, and hard bands) and the Lyapunov stability property to the final KER scores. For instance, one might prove that as corridors are tightened based on new empirical data, the probability of encountering a state outside the safe band decreases, which in turn increases the Knowledge-factor (K). Similarly, the integration of planes with high eco-impact, like carbon and biodiversity, should demonstrably raise the E score, while the confinement of risk within the Lyapunov residual should keep the R score bounded. The hex-stamped proofs produced as part of this research will serve as auditable, machine-checkable artifacts that bind these mathematical guarantees to specific lines of code and rows in the corridor tables, satisfying both internal development needs and external audit requirements [81]. This formal connection between the shared spine invariants and the KER gates is what makes the entire framework actionable and trustworthy for production deployment.

## Governance Architecture for Production Deployment Gates

The research outputs of empirically calibrated kernels and formally verified invariants are not intended for academic archives alone; they are explicitly designed to be directly integrated into a robust production governance architecture . This architecture serves as the bridge between research and operational deployment, ensuring that only rigorously vetted components are promoted to actuating roles. The system is structured around a series of gates, primarily governed by the `kerdeployable.aln` artifact, lane-based rules, and routing shells, which collectively enforce the safety and efficacy of the ecosafety stack . This creates a dual-form pattern where deliverables exist as both executable Rust/ALN artifacts for CI pipelines and as certifiable specifications for audits .

At the heart of the governance system is the `kerdeployable.aln` file, which acts as the top-level gatekeeper for all production activity . This artifact contains a set of formal invariants that must be satisfied for any given operation to proceed. These invariants include `invariant.corridorpresent` (ensuring all risk variables have defined corridor tables), `invariant.safestep` (guaranteeing that any proposed action does not increase the global residual $V_t$), and `invariant.kerdeployable` itself, which encapsulates the logic for passing the KER threshold checks . The production lanes require strict adherence to these invariants, enforcing minimum KER scores of K $\geqslant$ 0.90,

E ⩾ 0.90, and R ⩽ 0.13 . New risk coordinates from the five layers—`rFOG`, `rmaterials`, `rbiodiversity`, `rcarbon`, and `rcalib`—are simply treated as additional coordinates within the broader `RiskVector` that the core routing and actuation shell operates on . When a controller proposes a step, it receives a new risk vector (`rv_next`) and a residual value (`resid_next`). The system then invokes the `safestep` function, and only if the decision is "Accept" is the action passed on to the hardware actuators; otherwise, it is logged as a "Derate" or "Stop," preventing unsafe operations . This architecture ensures that the formal proofs of monotonicity and Lyapunov stability are continuously enforced at the lowest level of actuation.

A critical component of this governance model is the phased rollout of new or sharpened kernels, which is facilitated by a clear distinction between research and production environments. The default pathway for any new research kernel is to be deployed initially as a non-actuating diagnostic frame . Examples of such frames include `HydraulicDecayFrame`, `BiodiversityImpactFrame`, and `QualityMixingFrame`. These frames run continuously against historical data, such as replayed Phoenix shards, to generate time series of the new risk coordinates (`rx`) and the global residual (`Vt`) without having any effect on the physical system . This diagnostic mode allows researchers to validate the behavior of the new kernel in a wide variety of scenarios. The validation process involves running repeated shard replays to assess the kernel's performance. Only when the new kernel consistently demonstrates high KER scores—such as K > 0.94, E > 0.91, and R < 0.13 across diverse conditions—is it considered ready for actuation . This research phase is deliberately non-actuating, meaning that even if a proposed step would violate a newly-introduced corridor, the `safestep` function would log a "Derate" but not block the operation, allowing for safe observation and tuning . This iterative process of diagnosis, validation, and refinement ensures that only stable and effective kernels are ever considered for promotion.

Promotion from the research/diagnostic phase to the production phase is a deliberate and gated process. Initially, a new kernel would be enabled in an experimental lane (lane=EXP) . Even in this lane, it must still respect the `safestep` invariant, but the KER thresholds may be relaxed to allow for exploration. The `ResearchPlanShard` pilot program is designed to generate and track proposals for changing corridor definitions or introducing new kernels, providing a formal record of the evolution of the ecosafety specification . As the kernel matures and its performance is repeatedly validated, it can be promoted to a pre-production lane and eventually to a full production lane (lane=PROD). The promotion to a production lane triggers the enforcement of the strict KER gates (K⩾0.90, E⩾0.90, R⩽0.13) . The `FOG routing decision audit dataset`, which captures a large corpus of routing decisions with full predicate flags and downstream incident reports, serves as a final, empirical check on the system's behavior

under real loads . This dataset allows for the empirical validation of the `Vtnext ≤ Vt` invariants, the correctness of Accept/Reroute/Reject behaviors, and the impact of KER deltas in production-like scenarios . This entire governance architecture, from the `kerdeployable.aln` gates to the phased rollout and empirical auditing, is designed to make the transition from research to production as safe, transparent, and verifiable as possible, ensuring that the theoretical guarantees of the formal proofs translate into reliable, real-world safety.

## Ensuring Cross-Layer Coherence Through Unified Corridor Grammar

A primary challenge in designing a multi-layered system like the ecosafety stack is ensuring coherence and meaningful aggregation across domains that are fundamentally different in their physics, biology, and scale. The framework addresses this by establishing a unified semantic and statistical grammar for corridor definitions, while simultaneously allowing for flexibility in the underlying calibration methodologies. This dual approach ensures that disparate variables from the hydraulics, materials, biodiversity, carbon, and data quality planes can be meaningfully combined into a single, coherent risk metric, the global residual $V_t$, while still respecting the unique scientific basis of each domain .

The cornerstone of this cross-layer consistency is the adoption of a universal corridor schema, codified in the canonical `ecosafety.corridors.v2` definition . Every risk coordinate, regardless of its origin, must conform to this structure. This schema prescribes a set of standardized fields for each variable, including `safemin`, `safemax`, `goldmin`, `goldmax`, `hardmin`, and `hardmax` values, a weight `w_j`, a mandatory flag, and a Lyapunov channel designation . This uniform structure is what enables the aggregation formula $V_t = \sum_j w_j r_j^2$. Because every `r_j` is normalized to a common unit interval `[0,1]` through a simple, monotone mapping, the weighted sum of their squares forms a well-defined and comparable scalar residual . The weights `w_j` can be used to tune the sensitivity of the total residual to changes in different risk domains. The `mandatory` flag indicates whether a corridor must be present for a given operation, adding another layer of governance . This shared grammar effectively creates a lingua franca for risk, allowing the system to reason about trade-offs—for instance, whether a slight increase in hydraulic risk (`rhydraulics`) can be offset by a larger decrease in carbon risk (`rcarbon`).

While the grammar is rigidly unified, the methodologies used to calibrate the numeric endpoints of these corridors are permitted to be diverse and domain-specific . This methodological pluralism is a strength, as it allows each plane to leverage the most appropriate and robust evidence pipeline available to it, rather than forcing a one-size-fits-all statistical estimator . The key constraint is not the method itself, but the resulting property of the normalization function: it must be a monotone Lipschitz function in the direction of harm . This ensures that tightening a corridor boundary can only increase the corresponding `r_j` value and make meeting KER targets harder, never easier. The calibration *method* for each plane can therefore be tailored:

- **Hydraulics and FOG:** This plane can lean heavily on simulation and design standards. Corridor endpoints can be set by replaying historical Phoenix shards to find the boundaries of observed acceptable system performance or by using established hydraulic transport simulations to project outcomes under various loads [23] .
- **Materials:** Calibration for `rmaterials` is rooted in standardized laboratory testing. Kinetic parameters like t90 (time to 90% degradation) are derived from OECD 301 tests, toxicity metrics come from OECD 202, and leaching potential is measured according to ISO 14852 standards [25] [128] [129]. The resulting corridor bands (e.g., t90_safe ≤ 180 days, t90_hard ≤ 365 days) are direct translations of these lab results into risk coordinates .
- **Biodiversity:** Here, calibration relies on advanced spatial ecology models. Corridors for `rbiodiversity` are defined using connectivity metrics derived from circuit theory or least-cost path analysis, which are then calibrated with field survey data on patch importance and colonization rates [6] [7] . The normalization is designed so that degraded connectivity systematically pushes the `rbiodiversity` index toward 1 .
- **Carbon:** The `rcarbon` plane is calibrated based on the outputs of Life Cycle Assessments (LCA). Corridor thresholds are tied to lifecycle COe intensity values drawn from large, harmonized datasets like RASMI and are benchmarked against ISO 14044 standards [34] [36] [118].
- **Data Quality:** Finally, `rsigma` and `rcalib` are calibrated using metrology. Their corridor endpoints are not based on environmental data but on the known performance characteristics of sensors and data processing pipelines, guided by standards like ISO/IEC 17025 for measurement uncertainty and documented error rates for schema validation [1] [2] .

This approach allows for a sensible standard for calibration: default to pilot-backed quantiles where substantial field data exists (e.g., for HLR, PFAS, E. coli), and use simulation to test corridor robustness and shrink bands . For entirely new variables,

simulation-derived bands can be used provisionally but must be marked as experimental until validated by field data . This preserves a unified statistical *interpretation* at the corridor layer—bands correspond to either empirical quantiles of harm or scenario-based projections—while letting each plane use the evidence that best reflects its unique physics and ecology. The long-term plan explicitly involves "tightening corridors" using Phoenix shard replay, which implies a continuous, data-driven process of refining these empirical estimates while keeping the formal structure of the residual frozen . This balance between a fixed formal structure and flexible empirical calibration is what allows the system to be both robustly verifiable and adaptively responsive to new knowledge.

| Domain | Normalization Target | Calibration Method | Example Metric |
|---|---|---|---|
| **Hydraulics/ FOG** | Keep observed system in acceptable risk zones | Replayed historical shards; Design standards; Simulation [23] | `rsurcharge` (based on CEIM) |
| **Materials** | Translate lab results into risk coordinates | Standardized tests (ISO 14851, OECD 301/202); Lab kinetics; Field validation [25] [129] | `rt90` (days) |
| **Biodiversity** | Push index toward 1 with degraded connectivity | GIS models (Circuit Theory, LCP); Field surveys; Animal tracking [6] [7] | `rbiodiversity` (Connectivity Index) |
| **Carbon** | Map lifecycle intensity to risk coordinates | Harmonized LCA databases (RASMI); ISO 14044 standards [34] [36] | `rcarbon` (kg COe/kWh) |
| **Data Quality** | Quantify measurement and ingest error | Metrology (ISO/IEC 17025); GUM uncertainty propagation; Error injection experiments [1] [2] | `rcalib` (schema violation penalty) |

# Data Governance and Shard-Replayable Evidence Pipelines

A robust data governance framework is indispensable for the credibility and reproducibility of the ecosafety stack. The research framework mandates that every risk coordinate must be based on shard-replayable, CI-testable evidence, ensuring that all claims about system safety can be independently verified and that the calibration process is transparent and auditable . This is achieved through the establishment of a canonical shard schema, rigorous data migration procedures, and the use of hex-stamped proofs that bind mathematical guarantees to specific data and code artifacts. This focus on data lineage and provenance is critical for managing the complex evidence pipelines required for calibrating each of the five layers.

The foundation of the data governance structure is the canonical `qpudatashard` schema. The first step in this process is to conduct a census of all existing shard families

in use across the organization (e.g., Cyboquatic, EcoNet, AirGlobe, tray data, quantum data) and map their fields onto this proposed canonical schema . This mapping exercise produces a migration matrix that documents the transformations required to bring legacy data into the new, standardized format . The resulting `EcoNetSchemaShard2026v2.aln` becomes the authoritative source of truth for data structure, covering all real-world fields and not just theoretical ones . This standardization is the prerequisite for ensuring that data from disparate sources can be reliably ingested, processed, and aggregated into the `RiskVector` without ambiguity. It directly supports the "no schema, no write" principle, where deviations from the canonical schema are penalized through the `rcalib` coordinate, enforcing data integrity at the earliest stage of the pipeline .

To manage the transition from older schema versions to the new `v2` standard, a formal shard migration process is required. This involves designing and testing a migration pipeline using recorded datasets to ensure data loss and corruption are prevented . The output of this process is a set of verified migration recipes (old→new) accompanied by before-and-after KER comparisons. These recipes and their performance metrics become the normative examples provided in the official shard-migration tooling and documentation, creating a trusted and repeatable procedure for data engineers . This rigorous approach to data management extends to the calibration of the data quality plane itself. The `Ingest error / rcalib study` is a targeted research action designed to intentionally introduce controlled schema violations (e.g., missing columns, wrong data types, RFC-4180 breaks) and precisely measure their impact on the ingest process, the computed residual `V_t`, and the final KER score . The quantitative mapping generated from this study allows for the creation of explicit penalty functions for `rcalib`, turning data quality from a vague concern into a numerically tractable risk coordinate that directly influences system safety decisions .

The concept of "hex-stamping" is central to making the research outputs both executable and certifiable. A hex-stamp is a cryptographic hash or identifier embedded into key artifacts like Rust/ALN files, specification documents (`SPEC-ECOSAFETY.md`), and proofs . This practice creates an immutable link between a piece of code, a data calibration, and the formal proof that guarantees its safety properties. For example, a `Proof.rs` file implementing a kernel would be hex-stamped, and that same hex-stamp would appear in the `MetricField.aln` entry for the corresponding risk coordinate and in the text of the formal proof document . This creates a verifiable chain of custody: an auditor can take the hex-stamp, retrieve the exact version of the code, the exact version of the proof, and the exact calibration data (shard) used to instantiate the kernel, and replay the entire verification process. This dual-form pattern—executable kernels plus cert-ready documentation—is already applied to the existing planes and is being

extended to the five-layer decomposition . It transforms the system from a black box into a transparent, auditable artifact where safety claims are backed by concrete, replayable evidence.

Finally, the governance framework extends to the proposal and execution of changes through tooling like AI-chat and the `ResearchPlanShard` ALN grammar. The `researchplan-emitter` tool is used to generate real `ResearchPlanShards` for proposed corridor changes, which are then traced through the governance lanes . This creates a formal, auditable record of every modification to the ecosafety specification, complete with links to supporting evidence and the expected before-and-after KER scores . The AI-chat tooling is also put through "dry runs," where it is tasked with generating `ResearchPlanShards` and `SuggestionShards` against the frozen spec repository. The outputs of these runs are analyzed to create a map of potential failure modes, such as proposing unsafe or semantically invalid changes. These identified risks drive the development of additional ALN invariants and lane rules (e.g., restricting cross-plane dependencies, enforcing RESEARCH-only lanes) and provide concrete examples for training the chat tool to operate safely within the defined constraints . This comprehensive approach to data governance, combining schema standardization, rigorous migration, cryptographic provenance via hex-stamping, and formalized change management, ensures that the ecosafety stack is built on a foundation of trustworthy, verifiable, and resilient data.

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